

# Quantum Distortion Geometry and Nearest-Representative Classification of the Atypical Sector

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## Abstract

This note develops the third step of the structured quantum source-coding program. Starting from the atypical decomposition framework of Step 2, it introduces a distortion geometry for atypical states and uses that geometry to define nearest-representative classification rules. The main choice made here is to use the quantum Jensen–Shannon divergence as the primary symmetric comparison functional, with trace distance, fidelity, and relative entropy retained as secondary comparison tools. The guiding principle is that Step 3 should not merely describe an already chosen decomposition, but should generate a non-arbitrary one by assigning incoming atypical objects to the closest representative state. This yields a geometrically meaningful clustered decomposition of the atypical sector while preserving the asymptotic Schumacher baseline.

## 1 Introduction

Step 1 established the spectral typical–atypical split for a memoryless quantum source and identified the atypical sector as a low-weight spectral tail. Step 2 then introduced the formal bookkeeping of that tail: quantum probability zones, orthogonal atypical components, their weights, the dominant atypical weight, and the structural overhead associated with distinguishing components.

What Step 2 did *not* do was to explain how such components should be chosen. The decomposition remained intentionally abstract. Step 3 supplies the missing selection principle.

The role of the present step is therefore geometric rather than asymptotic. One wants a rule that says when two atypical objects should be regarded as similar, how representative states should be chosen, and how an incoming atypical sequence or state should be assigned to a structured class. In short, Step 3 must turn the formal decomposition of Step 2 into a geometry-generated decomposition.

The viewpoint adopted here is operationally simple. An incoming atypical object is assigned to the representative to which it is closest under a chosen quantum distance functional. The resulting classes are then precisely the clusters induced by that nearest-representative rule.

The main issue is therefore the choice of the comparison functional. Several candidates are natural:

- trace distance, because it controls operational distinguishability;
- fidelity, because it is tied to reconstruction quality;
- relative entropy, because it measures informational mismatch;
- quantum Jensen–Shannon divergence, because it is symmetric, finite, midpoint-based, and compatible with hierarchical coarse-graining.

The present note chooses the quantum Jensen–Shannon divergence as the primary geometric functional. This choice is motivated by three considerations. First, Step 3 is meant to induce rich internal structure rather than merely test one-shot distinguishability. Second, the midpoint

interpretation naturally supports the idea of representative states and hierarchical merging. Third, the local expansion of the Jensen–Shannon geometry reveals Fisher-information-type structure, making it a plausible entry point into a broader analytic geometry of the atypical sector.

The role of this step is therefore fivefold:

1. To define a quantum distortion geometry on atypical states.
2. To choose representative states for atypical classes.
3. To define nearest-representative classification of incoming atypical objects.
4. To induce a non-arbitrary clustered decomposition of the atypical sector.
5. To prepare the structured encoder of Step 4 by producing geometrically meaningful labels.

## 2 Quantum Distortion Functionals

Let  $\rho$  be the source state on  $\mathcal{H}$  and let  $\rho^{\otimes n}$  be the block source. Step 2 provides the atypical projector

$$Q_\epsilon^{(n)} = I - \Pi_\epsilon^{(n)}$$

and the atypical subspace

$$B_\epsilon^{(n)} = \text{Ran}(Q_\epsilon^{(n)}).$$

The present step requires a notion of geometric closeness on atypical objects. We therefore record the main comparison functionals that will be used.

**Definition 1** (Trace Distance). For density operators  $\tau$  and  $\omega$ ,

$$T(\tau, \omega) := \frac{1}{2} \|\tau - \omega\|_1.$$

**Definition 2** (Fidelity). For density operators  $\tau$  and  $\omega$ ,

$$F(\tau, \omega) := \left( \text{Tr} \sqrt{\sqrt{\tau} \omega \sqrt{\tau}} \right)^2.$$

**Definition 3** (Quantum Relative Entropy). For density operators  $\tau$  and  $\omega$ ,

$$D(\tau \| \omega) := \text{Tr}(\tau(\log_2 \tau - \log_2 \omega)),$$

whenever  $\text{supp}(\tau) \subseteq \text{supp}(\omega)$ , and  $D(\tau \| \omega) = +\infty$  otherwise.

**Definition 4** (Quantum Jensen–Shannon Divergence). For density operators  $\tau$  and  $\omega$ , define

$$\text{JSD}(\tau, \omega) := S\left(\frac{\tau + \omega}{2}\right) - \frac{1}{2}S(\tau) - \frac{1}{2}S(\omega).$$

The Jensen–Shannon divergence is the primary comparison functional in the present note. It may also be written as

$$\text{JSD}(\tau, \omega) = \frac{1}{2}D\left(\tau \left\| \frac{\tau + \omega}{2}\right.\right) + \frac{1}{2}D\left(\omega \left\| \frac{\tau + \omega}{2}\right.\right),$$

which makes explicit its midpoint interpretation:  $\text{JSD}(\tau, \omega)$  measures the entropy cost of forgetting which endpoint state was given and retaining only the average state.

**Remark 1.** The four functionals above do not play identical roles.

Trace distance is the cleanest operational distinguishability measure. Fidelity is most naturally interpreted as a reconstruction quantity. Relative entropy measures directional informational mismatch. The Jensen–Shannon divergence, by contrast, is the most natural midpoint-based functional and therefore the most suitable candidate for inducing representative-state clustering.

### 3 Why Jensen–Shannon Geometry is Chosen

The reason for preferring JSD at this stage is structural. Step 3 is not yet designing the final encoder and decoder. Its first responsibility is to induce the richest useful organization on the atypical sector. The Jensen–Shannon divergence is especially suitable for that role.

**Proposition 1** (Basic Properties of JSD). *For density operators  $\tau$  and  $\omega$ :*

- a)  $\text{JSD}(\tau, \omega) \geq 0$ ;
- b)  $\text{JSD}(\tau, \omega) = \text{JSD}(\omega, \tau)$ ;
- c)  $\text{JSD}(\tau, \omega) = 0$  if and only if  $\tau = \omega$ ;
- d)  $\text{JSD}(\tau, \omega)$  is always finite;
- e) if  $\tau$  and  $\omega$  are close, then the local second-order expansion of JSD yields a Fisher-information-type quadratic form.

*Proof.* Parts (a)–(d) are standard consequences of the concavity of the von Neumann entropy and the relative-entropy representation of JSD. Part (e) summarizes the local information-geometric fact that the Jensen–Shannon divergence has a quadratic leading behavior around coincident states, thereby inducing a local Fisher-type metric.  $\square$

The finiteness of JSD is particularly useful in the present setting. Relative entropy can become infinite under support mismatch, while fidelity and trace distance do not directly encode midpoint information. By contrast, JSD remains finite, symmetric, and explicitly tied to averaging. These are exactly the three properties one wants if a cluster representative is to stand for nearby atypical states in a controlled coarse-graining procedure.

**Remark 2.** The midpoint interpretation also suggests a hierarchy. Once representatives are formed, one may compare representatives to one another and form higher-level representatives again. This midpoint hierarchy is not developed fully here, but it motivates the present choice of geometry and points toward later multilevel refinements of AQIT.

### 4 Representative States and Incoming Atypical Seeds

To use any distance functional, one must specify both the incoming atypical object to be classified and the candidate representatives against which it is compared.

**Definition 5** (Atypical Spectral Word). Let

$$\rho = \sum_{i=1}^d \lambda_i |e_i\rangle \langle e_i|$$

and let

$$\mathbf{i} = (i_1, \dots, i_n)$$

be an atypical spectral word in the sense of Step 1, namely

$$\left| -\frac{1}{n} \log_2(\lambda_{i_1} \cdots \lambda_{i_n}) - S(\rho) \right| > \epsilon.$$

Its associated pure atypical seed state is

$$\tau_{\mathbf{i}}^{(n)} := |e_{\mathbf{i}}\rangle \langle e_{\mathbf{i}}|, \quad |e_{\mathbf{i}}\rangle := |e_{i_1}\rangle \otimes \cdots \otimes |e_{i_n}\rangle.$$

The present step works primarily with these atypical spectral seeds and with finite-blocklength coarse seeds built from them. The most important examples are:

- the pure atypical seed states  $\tau_{\mathbf{i}}^{(n)}$  themselves;
- normalized shell states in the diagonal-source example;
- normalized states supported on finite unions of atypical shells or zones.

Thus the incoming objects to be classified are not arbitrary density operators on  $B_\epsilon^{(n)}$ , but a specified seed family extracted from the Step 1 and Step 2 spectral structure.

**Definition 6** (Representative Family). A representative family for the atypical sector is a collection of density operators

$$\Omega^{(n)} = \{\omega_1^{(n)}, \dots, \omega_k^{(n)}\}$$

with each  $\omega_r^{(n)}$  supported in  $B_\epsilon^{(n)}$ .

The representative states need not initially coincide with the final components of Step 2. They are geometric seeds. Their purpose is to induce the final clustering through the distance-based assignment rule.

**Remark 3** (Representative Choice Principle). The usefulness of the induced clustering depends on the choice of representatives. A natural guiding principle is that a representative for a proposed seed cluster should minimize the expected within-cluster distortion. In the present Jensen–Shannon framework, this means that for a seed cluster  $\mathcal{S}_r^{(n)}$  with weights  $p_\tau$ , one seeks a state  $\omega_r^{(n)}$  that makes

$$\sum_{\tau \in \mathcal{S}_r^{(n)}} p_\tau \text{JSD}(\tau, \omega_r^{(n)})$$

small, ideally minimal within the admissible atypical sector.

The present step records this criterion only at the level of principle. The actual optimization of representatives, and the operational consequences of different choices, belong to Step 4.

**Remark 4.** This is the key conceptual shift from Step 2 to Step 3. Step 2 records the existence of a structured decomposition. Step 3 explains how such a decomposition can be generated. The components are not assumed first and justified later; rather, the representative geometry is chosen first, and the components are induced by nearest-representative classification of the atypical seeds.

## 5 Nearest-Representative Classification

We now formalize the clustering rule.

**Definition 7** (Nearest-Representative Label). Let  $d$  be a quantum comparison functional on density operators. Given a representative family

$$\Omega^{(n)} = \{\omega_1^{(n)}, \dots, \omega_k^{(n)}\}$$

and an atypical seed state  $\tau$ , define its nearest-representative label by

$$r^*(\tau) \in \arg \min_{1 \leq r \leq k} d(\tau, \omega_r^{(n)}).$$

**Definition 8** (Seed-Level Induced Clusters). Let  $\mathcal{S}_\epsilon^{(n)}$  denote the chosen family of atypical seed states. For each  $r \in \{1, \dots, k\}$ , define the induced seed cluster

$$:= \mathcal{S}_r^{(n)} := \left\{ \tau \in \mathcal{S}_\epsilon^{(n)} : r^*(\tau) = r \right\},$$

up to tie breaking on seeds with more than one minimizing representative.

Thus the clusters are the classes of atypical seeds attracted to the chosen representatives under the selected distortion geometry.

**Definition 9** (Jensen–Shannon Classification Rule). The primary Step 3 classification rule is obtained by taking

$$d(\tau, \omega) = \text{JSD}(\tau, \omega).$$

Then

$$r_{\text{JSD}}^*(\tau) \in \arg \min_{1 \leq r \leq k} \text{JSD}(\tau, \omega_r^{(n)}).$$

The importance of this rule is that it makes clustering midpoint-based. An incoming atypical seed is assigned to the representative whose average state with the seed incurs the smallest entropy gap.

**Remark 5.** This is the precise sense in which Step 3 gives geometry to the atypical decomposition. The geometry is not a passive label attached after the fact. It is the rule by which the label is assigned in the first place.

**Remark 6** (Cluster Cardinality as a Controlled Parameter). The number of representatives  $k$  is not fixed by geometry alone. A very coarse choice of  $k$  suppresses structure, while a very fine choice increases the Step 2 overhead parameter

$$\chi_Q = \frac{\log_2 k}{n(S(\rho) + \epsilon)}.$$

Accordingly, Step 3 should be read as generating admissible geometric clusterings rather than as selecting a unique value of  $k$ . Step 4 will impose the operational constraint that the fidelity gain produced by refined atypical treatment must justify the label overhead associated with the chosen  $k$ .

## 6 Consistency with the Step 2 Framework

The classification rule above must now be connected back to the Step 2 bookkeeping.

**Proposition 2** (Geometry-Induced Seed Partition). *Fix a representative family  $\Omega^{(n)}$  and a comparison functional  $d$ . Then the nearest-representative rule induces a partition of the incoming atypical state space into  $k$  classes, up to tie sets.*

*Proof.* By definition, each atypical seed  $\tau \in \mathcal{S}_\epsilon^{(n)}$  is assigned at least one minimizing index. The only ambiguity arises when several representatives attain the same minimal value. Away from such ties, the assignment is unique, and the induced classes partition the seed family.  $\square$

**Definition 10** (Geometry-Induced Component Subspaces). For each seed cluster  $\mathcal{S}_r^{(n)}$ , define the associated component subspace by

$$C_r^{(n)} := \text{span}\{\text{Ran}(\tau) : \tau \in \mathcal{S}_r^{(n)}\} \subseteq B_\epsilon^{(n)},$$

and let  $\Pi_r^{(n)}$  denote the projector onto  $C_r^{(n)}$ .

When the seeds are atypical spectral word states or shell states, the resulting subspaces are generated directly by the corresponding classified words or shells. In this way the Step 2 decomposition is recovered as the Hilbert-space realization of the Step 3 geometric partition.

**Proposition 3** (Weight Accounting Survives Classification). *Suppose the seed family consists of mutually orthogonal atypical seeds, such as atypical spectral word states or orthogonal shell states, and ties are resolved so that the seed clusters are disjoint. Then the component subspaces  $C_r^{(n)}$  are mutually orthogonal,*

$$Q_\epsilon^{(n)} = \sum_{r=1}^k \Pi_r^{(n)},$$

and the component weights

$$w_r^{(n)} := \text{Tr}(\rho^{\otimes n} \Pi_r^{(n)})$$

satisfy

$$\sum_{r=1}^k w_r^{(n)} = \text{Tr}(\rho^{\otimes n} Q_\epsilon^{(n)}) \leq \epsilon$$

for sufficiently large  $n$ .

*Proof.* If the seed clusters are disjoint and the seeds themselves are mutually orthogonal, then the spans generated by different clusters are mutually orthogonal as well. Since the union of the seed clusters exhausts the chosen atypical seed family, their spans exhaust the atypical subspace generated by that family. When the seed family is the full family of atypical spectral word states, this generated subspace is exactly  $B_\epsilon^{(n)}$ , hence

$$Q_\epsilon^{(n)} = \sum_{r=1}^k \Pi_r^{(n)}.$$

The weight identity then follows exactly as in Step 2. □

Thus Step 3 does not replace the bookkeeping of Step 2. It supplies the principle by which the bookkeeping objects are chosen.

## 7 Shells, Representatives, and the Diagonal Qubit Source

The diagonal qubit source remains the basic concrete example:

$$\rho = \text{diag}(1 - p, p), \quad 0 < p < 1.$$

As in Steps 1 and 2, the block source admits shell projectors

$$P_m^{(n)} = \sum_{\substack{\mathbf{i} \in \{1,2\}^n \\ m(\mathbf{i})=m}} |e_{\mathbf{i}}\rangle \langle e_{\mathbf{i}}|,$$

where  $m(\mathbf{i})$  is the number of occurrences of the second spectral symbol.

The normalized shell states

$$\sigma_m^{(n)} := \frac{P_m^{(n)} \rho^{\otimes n} P_m^{(n)}}{\text{Tr}(\rho^{\otimes n} P_m^{(n)})}$$

form a natural candidate family of incoming atypical objects or seed representatives.

**Proposition 4** (Shell-Based Seed Representatives). *For the diagonal qubit source, the atypical shells provide a canonical finite-blocklength family of seed states for nearest-representative classification.*

*Proof.* Each shell is spectrally well defined, carries a specific atypical weight, and is indexed by the empirical frequency  $m/n$ . Hence it is a natural finite-blocklength object to be used either as an incoming atypical state or as a representative from which larger clusters are built.  $\square$

This is the simplest place where JSD becomes geometrically useful. Shells with nearby empirical frequencies may be compared by their Jensen–Shannon divergence and merged if the divergence is sufficiently small. The resulting clusters are then no longer arbitrary shell unions; they are geometry-guided groups of empirically similar atypical shells.

More concretely, if  $\Delta_n > 0$  is a finite-blocklength merging threshold, then one may merge shells  $m$  and  $m'$  whenever

$$\text{JSD}\left(\sigma_m^{(n)}, \sigma_{m'}^{(n)}\right) \leq \Delta_n.$$

The threshold  $\Delta_n$  is not fixed in the present step. Its operational calibration is part of Step 4, where it must be balanced against the resulting cluster count and label overhead.

**Remark 7.** The shell example shows clearly why Step 3 is needed. Step 2 already permits every atypical shell to be a separate component. Step 3 asks which of those shells should be regarded as geometrically close enough to be grouped together under a controlled coarse-graining rule.

## 8 Secondary Comparison Functionals

Although JSD is the primary geometric functional, the other comparison quantities remain useful.

- **Trace distance.** Once clusters have been formed, trace distance can be used to test how operationally distinguishable neighboring representatives remain.
- **Fidelity.** Since Step 4 and Step 7 will ultimately care about reconstruction performance, fidelity remains a natural secondary diagnostic.
- **Relative entropy.** Relative entropy is especially useful when one wants to measure directional informational mismatch between an incoming atypical object and its assigned representative.

Accordingly, the present choice of JSD should not be read as excluding other metrics. Rather, it assigns roles: JSD induces structure, trace distance checks distinguishability, fidelity tracks reconstruction, and relative entropy measures directional mismatch.

**Remark 8.** The present note does not claim that the Jensen–Shannon geometry is globally optimal among all possible clustering rules. Its claim is narrower and sufficient for the present stage: JSD is a particularly natural structure-inducing choice because it is symmetric, finite, midpoint-based, and locally Fisher-geometric. The operational quality of that choice is to be judged in Step 4 by the rate–fidelity tradeoff it enables.

## 9 A Note on Orthogonality

For completeness, it is worth noting that the Step 2 decomposition is ultimately realized by orthogonal subspaces. One may therefore ask whether orthogonality trivializes the geometry introduced here.

The answer is no, because the primary Step 3 task is classification of incoming atypical seeds relative to representative states. The geometry is used to decide cluster membership, not to derive meaning solely from pairwise overlap of the final orthogonal subspaces.

**Remark 9.** If one compares fully normalized states supported on distinct final orthogonal components, some comparison functionals may indeed collapse to extreme values. This observation

does not invalidate the present construction. It merely shows that the geometric content of Step 3 lies in the assignment rule and the chosen representatives, rather than in the tautological fact that final orthogonal classes are perfectly separated after the classification has already been performed.

## 10 Toward Step 10: The Aryabhata Distance Functional

The Jensen–Shannon divergence is used here as the primary distortion functional because it has the right combination of symmetry, finiteness, midpoint meaning, and local Fisher-type expansion. At the same time, the broader AQIT program suggests a more ambitious possibility: that there exists a master analytic distance functional whose successive expansion layers recover several of the comparison structures currently used in the literature.

**Definition 11** (Programmatic Aryabhata Distance Functional). An *Aryabhata distance functional* is a proposed analytic quantum distance functional whose successive analytic layers are intended to recover increasingly refined geometric, entropic, and operational structures on quantum state space.

Associated with such a functional would be an *Aryabhata–Taylor expansion*, namely an analytic expansion whose successive layers organize the hidden comparison structure of the atypical sector and, where possible, recover known distance notions as derived levels, limiting cases, or descendants of a common framework.

The present note does not attempt to construct that master object. It merely supplies the first concrete geometric choice from which such a broader search may proceed. In this sense, JSD should be viewed both as the working metric of Step 3 and as a prototype for the analytic unification sought later in Step 10.

## 11 Forward Role of Step 3

The present step contributes the following ingredients to the larger ten-step program:

1. A concrete distortion geometry on atypical states.
2. Representative states for atypical classification.
3. A nearest-representative rule that induces non-arbitrary clusters.
4. A geometry-guided bridge from Step 2 bookkeeping to Step 4 encoder design.

What remains for Step 4 is to attach an explicit coding scheme to the resulting classes. Once the atypical sector has been partitioned into geometrically meaningful components, one can ask how much label overhead is required to distinguish them and what fidelity gain is achieved by treating them differently rather than discarding them uniformly.

What remains for Step 10 is different in character. There the aim is no longer merely to choose a useful working geometry, but to ask whether the geometric and operational quantities used throughout the program can be understood as layers of a common analytic distance framework. Step 3 therefore serves as the first concrete entry point into that later unification problem.

## 12 Conclusion

Step 3 supplies the first genuine geometry of the atypical quantum sector.

Instead of treating the Step 2 decomposition as an abstract partition, the present note introduces quantum distortion functionals and uses them to define nearest-representative classification of incoming atypical objects. The primary choice is the quantum Jensen–Shannon divergence, selected because it is symmetric, finite, midpoint-based, and locally Fisher-geometric. This



yields a non-arbitrary clustered decomposition of the atypical sector while preserving the Step 2 accounting framework and preparing the structured encoder of Step 4.

The central conceptual shift is therefore complete: the atypical sector is no longer merely decomposable in principle, but geometrically classifiable in practice.